

7.5.4 Rotations of Conics Worksheet

1. Identify and sketch the conic given by the equation $4x^2 - \sqrt{3}xy + 3y^2 - 5 = 0$

Since the equation contains an xy -term we know that it is a rotated conic.

We begin by attempting to find the angle of rotation.

$$\cot(2\theta) = \frac{4-3}{-\sqrt{3}} = \frac{1}{-\sqrt{3}} = -\frac{\sqrt{3}}{3} \Rightarrow 2\theta = \frac{2\pi}{3} \Leftrightarrow \theta = \frac{\pi}{3}$$

Now we begin to find the value of x and y :

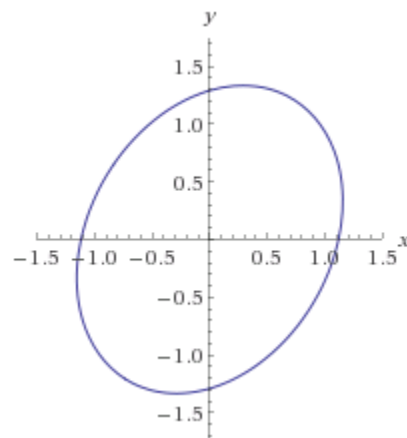
$$\begin{aligned} x &= x' \cos(\theta) - y' \sin(\theta) & x &= \frac{1}{2}x' - \frac{\sqrt{3}}{2}y' & x &= \frac{1}{2}(x' - \sqrt{3}y') \\ y &= x' \sin(\theta) + y' \cos(\theta) & y &= \frac{\sqrt{3}}{2}x' + \frac{1}{2}y' & y &= \frac{1}{2}(\sqrt{3}x' + y') \end{aligned}$$

We can use these equations to convert the given equation

$$\begin{aligned} 4\left(\frac{1}{2}(x' - \sqrt{3}y')\right)^2 - \frac{\sqrt{3}}{4}(x' - \sqrt{3}y')(\sqrt{3}x' + y') + 3\left(\frac{1}{2}(\sqrt{3}x' + y')\right)^2 - 5 &= 0 \\ \Leftrightarrow (x'^2 - 2\sqrt{3}x'y' + 3y'^2) - \frac{\sqrt{3}}{4}(\sqrt{3}x'^2 + x'y' - 3x'y' - \sqrt{3}y'^2) + \frac{3}{4}(3x'^2 + 2\sqrt{3}x'y' + y'^2) - 5 &= 0 \\ \Leftrightarrow \frac{19}{4}x'^2 + \frac{21}{4}y'^2 = 5 \\ \Leftrightarrow \frac{x'^2}{\left(\frac{20}{19}\right)} + \frac{y'^2}{\left(\frac{20}{21}\right)} = 1 \end{aligned}$$

This is an ellipse with a center of $(0,0)$. Note that $c^2 = \frac{20}{19} - \frac{20}{21} = \frac{40}{299} \Leftrightarrow c = \sqrt{\frac{40}{299}} \approx 0.3658$, so the

coordinates of the foci (with respect to the new axes) are $\left(\pm\sqrt{\frac{40}{299}}, 0\right)$.



2. Identify and sketch the conic given by the equation $2\sqrt{3}xy - 2y^2 - \sqrt{3}x - y = 0$

Since the equation contains an xy -term we know that it is a rotated conic.
We begin by attempting to find the angle of rotation.

$$\cot(2\theta) = \frac{0 - (-2)}{2\sqrt{3}} = \frac{\sqrt{3}}{3} \Rightarrow 2\theta = \frac{\pi}{3} \Leftrightarrow \theta = \frac{\pi}{6}$$

Now we begin to find the value of x and y :

$$\begin{aligned} x &= x' \cos(\theta) - y' \sin(\theta) \\ y &= x' \sin(\theta) + y' \cos(\theta) \end{aligned} \Leftrightarrow \begin{aligned} x &= \frac{\sqrt{3}}{2} x' - \frac{1}{2} y' \\ y &= \frac{1}{2} x' + \frac{\sqrt{3}}{2} y' \end{aligned} \Leftrightarrow \begin{aligned} x &= \frac{1}{2} (\sqrt{3} x' - y') \\ y &= \frac{1}{2} (x' + \sqrt{3} y') \end{aligned}$$

We can use these equations to convert the given equation

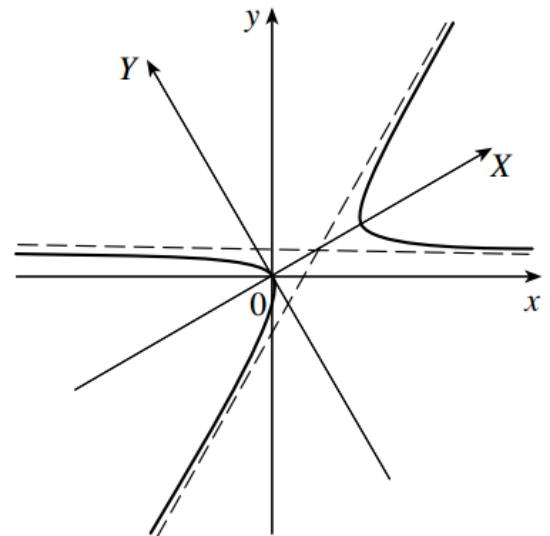
$$\begin{aligned} 2\sqrt{3} \frac{1}{4} (\sqrt{3} x' - y') (x' + \sqrt{3} y') - 2 \left(\frac{1}{2} (x' + \sqrt{3} y') \right)^2 - \frac{\sqrt{3}}{2} (\sqrt{3} x' - y') - \frac{1}{2} (x' + \sqrt{3} y') &= 0 \\ \Leftrightarrow \frac{\sqrt{3}}{2} (\sqrt{3} x'^2 + 2x'y' - \sqrt{3} y'^2) - \frac{1}{2} (x'^2 + 2\sqrt{3} x'y' + 3y'^2) - \frac{\sqrt{3}}{2} (\sqrt{3} x' - y') - \frac{1}{2} (x' + \sqrt{3} y') &= 0 \\ \Leftrightarrow x'^2 - 3y'^2 - 2x' &= 0 \\ \Leftrightarrow (x'^2 - 2x' + 1) - 3y'^2 &= 1 \\ \Leftrightarrow (x' - 1)^2 - 3y'^2 &= 1 \\ \Leftrightarrow \frac{(x' - 1)^2}{1} - \frac{y'^2}{(1/3)} &= 1 \end{aligned}$$

This is a hyperbola with a center of $(1, 0)$. Note that $c^2 = 1 + \frac{1}{3} = \frac{4}{3} \Leftrightarrow c = \sqrt{\frac{4}{3}} = \frac{2\sqrt{3}}{3}$, so the coordinates

of the foci (with respect to the new axes) are $\left(1 \pm \frac{2\sqrt{3}}{3}, 0\right)$.

Also note that the equations for the asymptotes with respect to the new axes are:

$$y' = \pm \frac{\sqrt{(1/3)}}{1} (x' - 1) \Leftrightarrow y' = \pm \frac{1}{\sqrt{3}} (x' - 1)$$



3. Show that the given equation is a parabola, then find its vertex, focus, and directrix in terms of the xy -axes and sketch a graph.

$$6x^2 + 4\sqrt{3}xy + 2y^2 - 9x + 9\sqrt{3}y - 63 = 0$$

Since the equation contains an xy -term we know that it is a rotated conic.

We begin by attempting to find the angle of rotation using the previous theorem.

$$\cot(2\theta) = \frac{6-2}{4\sqrt{3}} = \frac{1}{\sqrt{3}} \Leftrightarrow 2\theta = \frac{\pi}{3} \Leftrightarrow \theta = \frac{\pi}{6}$$

Since this equation is not easy to solve for θ we will instead use a triangle and some trig. identities:

$$\begin{aligned} x &= x' \cos(\theta) - y' \sin(\theta) & \Leftrightarrow & \quad x = \frac{\sqrt{3}}{2}x' - \frac{1}{2}y' & \quad x = \frac{1}{2}(\sqrt{3}x' - y') \\ y &= x' \sin(\theta) + y' \cos(\theta) & \Leftrightarrow & \quad y = \frac{1}{2}x' + \frac{\sqrt{3}}{2}y' & \quad y = \frac{1}{2}(x' + \sqrt{3}y') \end{aligned}$$

We can use these equations to convert the given equation

$$\begin{aligned} &\frac{6}{4}(\sqrt{3}x' - y')^2 + \sqrt{3}(\sqrt{3}x' - y')(x' + \sqrt{3}y') + \frac{1}{2}(x' + \sqrt{3}y')^2 - \frac{9}{2}(\sqrt{3}x' - y') + \frac{9\sqrt{3}}{2}(x' + \sqrt{3}y') - 63 = 0 \\ \Leftrightarrow &\frac{3}{2}(3x'^2 - 2\sqrt{3}x'y' + y'^2) + \sqrt{3}(\sqrt{3}x'^2 + 2x'y' - \sqrt{3}y'^2) + \frac{1}{2}(x'^2 + 2\sqrt{3}x'y' + 3y'^2) - \frac{9}{2}(\sqrt{3}x' - y') + \frac{9\sqrt{3}}{2}(x' + \sqrt{3}y') - 63 = 0 \\ \Leftrightarrow &8x'^2 + 18y' - 63 = 0 \\ \Leftrightarrow &18y' = -8x'^2 + 63 \\ \Leftrightarrow &y' = -\frac{8}{18}x'^2 + \frac{63}{18} \\ \Leftrightarrow &y' = -\frac{4}{9}x'^2 + \frac{7}{2} \end{aligned}$$

This is a parabola with $a = \frac{1}{4p} = -\frac{4}{9} \Leftrightarrow p = -\frac{9}{16}$, $h = 0$, & $k = \frac{7}{2}$.

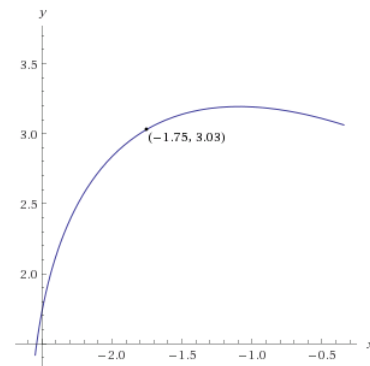
Therefore it has vertex $\left(0, \frac{7}{2}\right)$, focus $\left(0, -\frac{7}{2} + \frac{9}{16}\right) = \left(0, -\frac{47}{16}\right)$, and directrix $y' = -\frac{7}{2} - \frac{9}{16} = -\frac{65}{16}$ with respect to the $x'y'$ -axes.

To convert these to points/lines in the xy axes we will use the equations from the first theorem:

$$\begin{aligned} \text{Vertex: } &x = 0 \cdot \cos\left(\frac{\pi}{6}\right) - \frac{7}{2} \sin\left(\frac{\pi}{6}\right) = -\frac{7}{4} \\ &y = 0 \cdot \sin\left(\frac{\pi}{6}\right) + \frac{7}{2} \cos\left(\frac{\pi}{6}\right) = \frac{7\sqrt{3}}{4} \end{aligned} \rightarrow \left(-\frac{7}{4}, \frac{7\sqrt{3}}{4}\right)$$

Focus:

$$\begin{aligned} x &= 0 \cdot \cos\left(\frac{\pi}{6}\right) - \frac{47}{16} \sin\left(\frac{\pi}{6}\right) = -\frac{47}{32} \\ y &= 0 \cdot \sin\left(\frac{\pi}{6}\right) + \frac{47}{16} \cos\left(\frac{\pi}{6}\right) = \frac{47\sqrt{3}}{32} \end{aligned} \rightarrow \left(-\frac{47}{32}, \frac{47\sqrt{3}}{32}\right)$$



Directrix: We have $y' = -\frac{65}{16}$, so

$$y' = -x \sin\left(\frac{\pi}{6}\right) + y \cos\left(\frac{\pi}{6}\right) \Leftrightarrow -\frac{65}{16} = -\frac{1}{2}x + \frac{\sqrt{3}}{2}y \Leftrightarrow y = \frac{1}{\sqrt{3}}x - \frac{65}{8\sqrt{3}}$$

4. Identify and sketch the conic given by the equation $153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0$

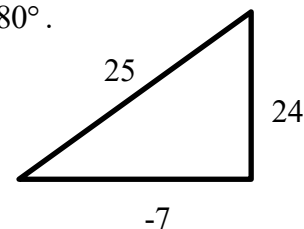
Since the equation contains an xy -term we know that it is a rotated conic.

We begin by attempting to find the angle of rotation using the previous theorem.

$$\cot(2\theta) = \frac{153 - 97}{-192} = -\frac{56}{192} = -\frac{7}{24}$$

Since this equation is not easy to solve for θ we will instead use a triangle and some trig. identities. We should note that since we can always select θ as an acute angle this implies that $2\theta < 180^\circ$.

Therefore since $\cot(2\theta) < 0$ this implies that $\cos(2\theta) < 0$ & $\sin(2\theta) > 0$.



So now we calculate:

$$\begin{aligned} \cos^2(\theta) &= \frac{1}{2}[1 + \cos(2\theta)] & \Leftrightarrow & \cos^2(\theta) = \frac{1}{2}\left[1 - \frac{7}{25}\right] & \Leftrightarrow & \cos^2(\theta) = \frac{18}{50} & \Leftrightarrow & \cos(\theta) = \frac{3}{5} \\ \sin^2(\theta) &= \frac{1}{2}[1 - \cos(2\theta)] & \Leftrightarrow & \sin^2(\theta) = \frac{1}{2}\left[1 + \frac{7}{25}\right] & \Leftrightarrow & \sin^2(\theta) = \frac{32}{50} & \Leftrightarrow & \sin(\theta) = \frac{4}{5} \end{aligned}$$

Notice that now we do not need to find the value of θ and can begin to find the value of x and y :

$$\begin{aligned} x &= x' \cos(\theta) - y' \sin(\theta) & \Leftrightarrow & x = \frac{3}{5}x' - \frac{4}{5}y' & \Leftrightarrow & x = \frac{1}{5}(3x' - 4y') \\ y &= x' \sin(\theta) + y' \cos(\theta) & \Leftrightarrow & y = \frac{4}{5}x' + \frac{3}{5}y' & \Leftrightarrow & y = \frac{1}{5}(4x' + 3y') \end{aligned}$$

We can use these equations to convert the given equation

$$153\left(\frac{1}{5}(3x' - 4y')\right)^2 - 192\frac{1}{25}(3x' - 4y')(4x' + 3y') + 97\left(\frac{1}{5}(4x' + 3y')\right)^2 - 30\frac{1}{5}(3x' - 4y') - 40\frac{1}{5}(4x' + 3y') - 200 = 0$$

$$\Leftrightarrow \frac{153}{25}(9x'^2 - 24x'y' + 16y'^2) - \frac{192}{25}(12x'^2 - 7x'y' - 12y'^2) + \frac{97}{25}(16x'^2 + 24x'y' + 9y'^2) - (18x' - 24y') - (32x' + 24y') = 200$$

$$\Leftrightarrow 153(9x'^2 - 24x'y' + 16y'^2) - 192(12x'^2 - 7x'y' - 12y'^2) + 97(16x'^2 + 24x'y' + 9y'^2) - 25(18x' - 24y') - 25(32x' + 24y') = 5000$$

$$\Leftrightarrow 1377x'^2 - 3672x'y' + 2448y'^2 - 2304x'^2 + 1344x'y' + 2304y'^2 + 1552x'^2 + 2328x'y' + 873y'^2 - 450x' + 600y' - 800x' - 600y' = 5000$$

$$\Leftrightarrow 625x'^2 + 5625y'^2 - 1250x' = 5000$$

$$\Leftrightarrow x'^2 + 9y'^2 - 2x' = 8$$

$$\Leftrightarrow (x'^2 - 2x' + \quad) + 9y'^2 = 8$$

$$\Leftrightarrow (x'^2 - 2x' + 1) + 9y'^2 = 8 + 1$$

$$\Leftrightarrow \frac{(x' - 1)^2}{9} + \frac{y'^2}{1} = 1$$

This is an ellipse with a center of $(1, 0)$. Note that $c^2 = 9 - 1 = 8 \Leftrightarrow c = 2\sqrt{2}$, so the coordinates of the foci (with respect to the new axes) are $(\pm 2\sqrt{2}, 0)$.

